Aakash Kulkarni

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SUMMARY

Software Engineer at Cloudvik with 2+ years of experience building reliable backend and cloud systems, with a focus on GenAI tooling. AWS Certified Solutions Architect – Associate. I care about software that meets real user needs.

SKILLS

Programming: Java, Python, C++, JavaScript, SQL, Bash

Frameworks: Spring Boot, Node.js, FastAPI, React, Kafka

Cloud & DevOps: AWS, GCP, Docker, Kubernetes, Terraform, CI/CD

Data & AI: PostgreSQL, MySQL, MongoDB, LangChain, RAG, MCP, PyTorch

EXPERIENCE

Software Engineer
Sep 2025 – Present

Cloudvik — *Remote*

- Building Spring Boot microservices that orchestrate LangChain extraction pipelines for enterprise document processing.
- Developing REST APIs for a sensor integration platform that ingests real-time telemetry and syncs it with cloud infrastructure.
- Leveraging an AI coding harness and MCP tools to accelerate implementation, reviews, and documentation; applying AI governance through scoped tool access, human review gates, and auditable changes.

Graduate Teaching Assistant
Mar 2025 – Jun 2025

Oregon State University — *Corvallis, OR*

- Supported a group of over 100 students in Parallel Programming, helping them navigate concepts like multi-threading and concurrent patterns.

Graduate Research Assistant
Apr 2024 – Mar 2025

Oregon State University — *Corvallis, OR*

- Participated in the development of HeteroBugDetect, a framework designed to identify platform-specific bugs in scientific software (LAMMPS).
- Assisted in coordinating automated testing workflows on distributed HPC systems to improve the efficiency of identifying software vulnerabilities.

Associate Software Developer
Jun 2021 – Mar 2023

Zemoso Technologies — *Hyderabad, India*

- Contributed to a team building a financial data dashboard for TransUnion that provided real-time visualization of credit trends and risk metrics for enterprise users.
- Helped optimize database structures for TransUnion's credit record management, which improved data retrieval speeds for financial modules.
- Assisted in implementing secure OAuth 2.0 authentication workflows to protect sensitive consumer financial data and maintain compliance standards.

Mobile Application Developer Intern
Jul 2021 – Sep 2021

FindMind Analytics — *Remote*

- Developed ZuPay, a mobile banking app for teenagers with parental controls, using Flutter for cross-platform iOS and Android.
- Launched on the iOS and Android app stores; the app reached 10,000+ active users.

SELECTED PROJECTS

JAAABS: Job Application Automator & Aakash's Bot Studio | *Python, Playwright, LangChain, Next.js, SQLite* 2026

- Built a human-in-the-loop job application studio that discovers companies, scores roles, tailors a LaTeX resume per position, and autofills applications in Chromium.
- Designed human safety gates for login, CAPTCHA, OTP, review, and submit, with an evidence ledger and a local dashboard.

AWS Infrastructure Automation | *Terraform, AWS, Python, Bash* Jan 2025

- Built reusable Terraform modules for VPC, subnets, security groups, EC2, RDS, S3, and Route 53 across dev, staging, and production.
- Automated deployment and monitoring with Python and Bash, and documented runbooks for scaling and connectivity issues.

EDUCATION

Oregon State University Corvallis, OR
Master of Science in Computer Science (GPA: 3.91/4.0) *Mar 2023 – Jun 2025*
Advisor: Dr. Manish Motwani. Relevant coursework: High Performance Computing, Deep Learning, Software Engineering Methodologies, Parallel Programming, Mobile Application Development.

Jawaharlal Nehru Technological University Hyderabad, India
Bachelor of Technology in Computer Science (GPA: 7.25/10.0) *Jul 2018 – Aug 2022*
Relevant coursework: Data Structures, Algorithms, Operating Systems, Database Management Systems, Computer Networks, Distributed Systems.

RESEARCH

LLM-Guided Differential Fuzzing for Detecting Platform-Specific Bugs in Scientific Applications — IEEE AIxSE 2025 (co-authored). Paper.
Co-developed HeteroBugDetect; detected 8 of 20 known bugs and 2 previously unknown bugs in LAMMPS.

CERTIFICATIONS, ACHIEVEMENTS & INTERESTS

AWS Certified Solutions Architect – Associate (Jan 2026).
1st Place, OSU Winter Hackathon 2024 — Guitar Hub, an AI-powered guitar tuner and song analysis tool using OpenAI and Hugging Face.
Video games, cooking, strength training, and desktop ricing.